

SANJAY SENTHIL

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PROFESSIONAL SUMMARY

Results-driven AI & Data Science engineer with deep expertise in machine learning, computer vision, and scalable data engineering. Skilled in building end-to-end ML pipelines, deploying deep learning models, and working across the full data lifecycle — from ingestion to inference. Proficient in Python, PyTorch, Apache Spark, and cloud platforms, with a strong foundation in both research and production-grade implementation. A fast learner with a problem-solving mindset, passionate about leveraging AI to create meaningful, real-world impact across industries.

EDUCATION

Amrita Vishwa Vidyapeetham

Coimbatore, India

Bachelor of Technology in Artificial Intelligence and Data Science

Aug 2023 – Jul 2027

- Collaborated on academic lab projects, implementing algorithms for deep learning, NLP, and computer vision from scratch using PyTorch and scikit-learn.
- Participated in competitive hackathons and department symposiums, representing the university in algorithm design and machine learning contests.

EXPERIENCE

Cognifyz Technologies

Remote

Data Science Intern

Dec 2025 – Jan 2026

- Engineered a restaurant rating prediction regression model (Random Forest, Gradient Boosting) using scikit-learn, achieving a Test R^2 of 0.52 and RMSE of 0.68, while eliminating data leakage by removing post-facto variables.
- Developed a location-intelligent content-based recommendation system using a multi-criteria ranking algorithm (weighted by rating, popularity, budget) and the Haversine formula to compute distance.
- Conducted location-based exploratory data analysis (EDA) across 15 cities, segmenting pricing zones and identifying a 27% online delivery service gap to advise expansion strategies.

PROJECTS

Real-Time Multimodal Emotion Detection

[GitHub](#)

YOLO, Vision Transformer (ViT), Scala, Spark Streaming, HDFS, Python

- Built a real-time multimodal emotion detection system achieving **30+ FPS** for **9-class** facial expression recognition using a hybrid **YOLO + Vision Transformer (ViT)** architecture.
- Improved data ingestion throughput by **8.5×** by designing a **Spark Structured Streaming Medallion pipeline** (Bronze/Silver/Gold) on HDFS and converting raw images to Snappy Parquet using Scala.

Music Emotion Recognition using CNN and HuBERT

[GitHub](#)

PyTorch, HuBERT, Librosa, Python

- Developed an end-to-end music emotion recognition system in **PyTorch**, achieving **89.83% test accuracy** and **93.62% validation accuracy** on an **8-class** Valence-Arousal dataset containing **10,812 samples**.
- Enhanced model generalization by **14%** through a hybrid **CNN-HuBERT** architecture combining MFCC features with pretrained speech embeddings and data augmentation techniques.

Monte Carlo Simulation in Supply Chain Management

[GitHub](#)

C++, OpenMPI, Python, Terraform, Azure, scikit-learn

- Developed a high-performance **Monte Carlo simulation engine** in C++ to model stochastic inventory dynamics across **52 planning periods**, reducing forecasted inventory costs by **18%**.
- Accelerated execution by **15×** for **100,000+ simulations** by parallelizing workloads across a multi-node **Azure VM cluster** using OpenMPI and provisioning infrastructure with Terraform.

TECHNICAL SKILLS

Programming Languages: Python, C++, Scala, C, SQL

Libraries & Frameworks: PyTorch, scikit-learn, Spark Streaming, OpenCV, Pandas, NumPy, Hugging Face (Transformers), Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, VS Code, Terraform, Microsoft Azure, Hadoop, OpenMP, MPI, Apache Kafka, Linux/Unix Shell

ACHIEVEMENTS & CERTIFICATIONS

- **IEEE YESIST12 2026 Finalist** – Selected among **2,000+ global submissions** for the final round of the Innovation Challenge for the project *Quantum-Enabled Capacity Expansion Planning for Resilient Green Grids*.
- **Career Essentials in Generative AI** – Microsoft & LinkedIn Learning (Completed Nov 2025) | [Verify Credential](#)
- **Build Your Generative AI Productivity Skills** – Microsoft & LinkedIn Learning (Completed Nov 2025) | [Verify Credential](#)